

# Meaning after the Multimodal Revolution: Seven Things Signs and Gestures Teach Us about Meaning\*

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**Abstract.** Following a "multimodal revolution" in the language sciences, meaning is now studied not just in speech but also in sign languages ('sign') and in gestures. Seven lessons can be drawn from recent research. (1) Sometimes sign makes visible key logical elements that are covert in speech. (2) Sign combines the same meaning operations as speech with a richer depictive component. (3) Sign combines the same kind of syntax as speech with a dedicated syntax for its depictive elements. (4) Even the purely depictive component of sign is linguistically structured. (5) Gestures that replace words share some non-trivial properties with signs—but unlike signs, they are not words. (6) Gestures that accompany words make backgrounded (non-assertive) contributions. (7) Several of these properties apply beyond signs and gestures, suggesting that they have a deeper cognitive source, not a narrowly linguistic one. These lessons show that human meaning results from both discrete, logical-like and gradient, iconic components, integrated through powerful cognitive mechanisms across sign, speech and gestures.

Keywords: multimodality, sign language, gestures, semantics, iconicity

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## Introduction: The multimodal revolution in linguistics

Semantics, the study of meaning, used to be a theory of meaning *in speech*. In the last 20 years, this overly narrow focus has been broadened to encompass sign languages and gestures<sup>1</sup> following what is sometimes called the "multimodal revolution" in linguistics.<sup>2</sup> We propose to draw its main lessons for our understanding of meaning, with special attention to theories coming out of formal semantics.

To provide some historical context, formal linguistics arose in the 1960s (for syntax and phonology) and in the 1970s (for semantics) from the attempt of constructing fully explicit models of language. Formal linguistics started investigating sign languages early on, starting in the 1960 for the articulation of signs (sign language phonology; Stokoe 1960, Klima and Bellugi 1979; see van der Hulst 2022), and in the 1970s for the study of rules by which signs can be combined into sentences (sign language syntax; Fischer 1975, Klima and Bellugi 1979). By contrast, the formal analysis of sign language meaning lagged behind and only developed in the 2000s (Zucchi 2004, 2012). It built on sign language syntax and, later, on detailed studies of sign language meaning in other traditions, notably cognitive linguistics (e.g. Liddell 2003, Taub 2001, Dudis 2004, Janzen 2025).

The formal semantics of gestures developed in two waves. The first wave focused on the interaction between gestures and discourse structure, within frameworks that had been developed to handle discourse-level phenomena, and with a focus on attested (corpus-based) examples (Lascarides and Stone 2009). The second wave specifically asked how sentence-level grammar interacts with gestures, and made systematic use of constructed examples to test detailed predictions, hence powerful new insights (Ebert and Ebert 2014, Ebert 2024).

Signs are not gestures. Throughout their history, sign languages have suffered from the entirely erroneous view that they are just gestural systems with no grammar and severe expressive limitations.<sup>3</sup> Any experience with sign language refutes this view. Science does too: both linguistic and neurological studies have shown that human language exists with similar grammatical, expressive and brain properties in the vocal and in the visual modality (Sandler and Lillo-Martin 2006, Lillo-Martin and Gajewski 2014, MacSweeney et al. 2008; it is still the case that each modality has its specificities, to the point that historically unrelated sign languages function as a "language family" for purposes of typological work [e.g. Sandler and Lillo-Martin 2006, Lillo-Martin and Gajewski 2014]).

Sign language linguistics initially devoted much attention to the discrete and grammatical component of signs rather than to their gradient and iconic component, in part to counter the view that signs are just gestures. But the linguistic status of sign languages is now so strongly established that recent research has investigated the relation between speech, sign and gestures (Goldin-Meadow and Brentari 2017)—with important new insights, as we will now see.

### Lesson 1. Sign makes visible certain elements of the logical structure of sentences that are covert in speech.

In view of the general linguistic and neurological similarities between sign and speech, it is unsurprising that the main elements of meaning are common to both (Sandler and Lillo-Martin 2006, Schlenker 2018a). For instance, just as English has rich means to count objects, expressions like *every*, *some*, and *most*, ASL has related expressions. In English, pronouns can play the role of logical variables. For instance, *[Every man]<sub>x</sub> knows that he<sub>x</sub> is mortal* approximates the logical formula *For all x, if x is a man, x is mortal* (or in logical notation:  $\forall x (\text{man}(x) \Rightarrow \text{mortal}(x))$ ). Pronouns with similar function exist

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<sup>1</sup> While one sometimes thinks that there is a single sign language across countries, this is not so. Sign languages arise spontaneously just as spoken languages do, and are correspondingly diverse, with properties that needn't be borrowed from the surrounding spoken languages. As an example (Geraci 2004), Italian Sign Language (LIS) has the basic word order Subject-Object-Verb, whereas Italian goes with Subject-Verb-Object. For things one should know about sign language, see Emmorey 2023. For further information about sign language creation, see Brentari and Coppola 2013. For a different take on gestures and on the gestures/signs connection, see Özyürek 2012, Özyürek and Woll 2019.

<sup>2</sup>The term "multimodal turn" has been used as well (thanks to A. Özyürek for discussion).

<sup>3</sup> On the importance of giving deaf children access to a sign language, see for instance Napoli et al. 2015.

in sign as well, typically realized by way of pointing—towards the signer or addressee for the first or second person, towards other positions for third person.<sup>4</sup>

But a striking discovery was that some key logical components that are invisible in speech are sometimes overtly realized in sign (Lillo-Martin and Klima 1990, Schlenker 2018a). Such is the case of variables. In English, *Sarkozy told Obama that he would win the election* is ambiguous, as *he* can refer to Sarkozy, or to Obama. Linguists have traditionally analyzed this ambiguity by positing that the sentence can be given two different cognitive representations, shown here:

Sarkozy<sub>x</sub> told Obama<sub>y</sub> that **he<sub>x</sub>** would win the election.

Sarkozy<sub>x</sub> told Obama<sub>y</sub> that **he<sub>y</sub>** would win the election.

In the first representation, *he* carries the same variable *x* as *Sarkozy*; in the second, *he* carries the same variable *y* as *Obama*. But these variables are postulated on indirect grounds, as they are not overtly realized in speech. Strikingly, these variables can be overtly realized in sign: one starts by assigning a position or "locus" *x* (which just means *place* in Latin) to Sarkozy, and another locus *y* to Obama. Depending on whether the pronoun points towards *x* or *y*, two different meanings are obtained: each locus plays the role of a variable.

This finding of "Logical Visibility" has been extended to several other key components of the logical structure of the sentences (Schlenker 2018a). As a result, sign languages may help decide certain foundational theoretical issues thanks to the overtness they sometimes (not always) bring to the table.

## Lesson 2. Sign combines the same kind of semantics as speech with a richer depictive component.

Vocal iconicity exists in spoken language (e.g. Dingemanse et al. 2015, Lockwood and Dingemanse 2015), for instance when one says that *the talk was looong* to mean that it was very long. An expression is iconic if it refers to something by virtue of resembling it (the length of the vowel resembles the duration of the talk), or more formally if there is a structure-preserving map between the expression and what it refers to (the longer the vowel, the longer the talk). Iconic effects are richer and more pervasive in sign than in speech (they are often called "depictive" because they involve visual iconicity). In an ASL version of *my group grew*, the verb can be realized in at least six different ways, as illustrated below (Schlenker 2018a): the broader the sign's endpoints, the larger the growth; the quicker the movement, the faster the grow (video: <https://youtu.be/QOSBBSZCGVU>)

Figure 1. Different iconic modifications of the sign *GROW* in ASL

|                      | Narrow endpoints                                                                    | Medium endpoints                                                                     | Broad endpoints                                                                       |
|----------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                      |  |  |  |
| <b>Slow movement</b> | small amount, slowly                                                                | medium amount, slowly                                                                | large amount, slowly                                                                  |
| <b>Fast movement</b> | small amount, quickly                                                               | medium amount, quickly                                                               | large amount, quickly                                                                 |

These iconic effects are not restricted to the concrete vocabulary (e.g. Lillo-Martin and Gajewski 2014). For instance, "reflect" may be realized with a longer duration to refer to a longer reflection, or at an accelerating speed if the reflection intensified.

Strikingly, iconic effects are also found in key logical components of sign language, such as pronouns. If an ASL or LSF (French Sign Language) signer talks about a very tall man training to be

<sup>4</sup> Combined, these two properties—the existence of quantifiers, and that of variables—gave rise to rich studies of the expressive power of human languages, which go far beyond standard logic (Keenan 1996; see Schlenker 2010 for a brief survey). In this respect, sign seems similar to speech.

an astronaut, they may point upwards because this is where the man's head is. But if they recount that as part of his training the man was rotated and ended up in upside down position, they will point downwards instead (Schlenker 2018a; Liddell 2003). In this case, the position one points to is simultaneously a logical variable (as argued above) and a simplified picture of what it represents.

In *grow* and in pronouns, iconic modulation is optional. But across sign languages, there are other expressions, called classifiers, which are iconic almost by definition (Emmorey and Herzig 2003, Zwitserlood 2012, Dudis 2004, Huenerfauth 2004). Classifiers have a conventional manual shape just like any word, as illustrated in Fig. 2. But their position and movement are entirely free (unlike those of other signs), and interpreted iconically. Thus the upright index of Fig. 2b can move rightward if that is the person's movement, slowly if the person walks slowly, etc.

Figure 2.

a. A classifier representing a vehicle



b. A classifier representing an upright elongated object or an upright person



Credits: ©Dr. Bill Vicars, <https://www.lifeprint.com/asl101/pages-signs/classifiers/classifiers-main.htm>

In sum, sign languages offer an extraordinary synthesis of two interacting systems: a discrete component with the same logical structure as speech, and a much richer iconic component.

### Lesson 3. Sign combines the same kind of syntax as speech with a dedicated syntax for its depictive elements.

Different sign languages have different word orders, just as is the case for spoken languages. But a surprising finding is that classifiers often override this word order (Pavlič 2016): alongside their iconic semantics, sign languages also have an iconic syntax, in the sense that classifiers take their arguments in the order in which these would be represented in a simplified comic.

Take the sentence *A whale swallowed a person*. The basic word order of ASL is SVO (Subject-Verb-Object), so with the normal verb *EAT* the sentence is: WHALE EAT PERSON. But one can also tell the story with a classifier representing a whale, moving and opening its mouth to eat up a person (transcribed as *SWIM-SWALLOW-cl*). Strikingly, in this case the preferred word order has the object before the verb, for instance with the order SOV (Subject-Object-Verb): WHALE PERSON SWIM-SWALLOW-cl (see Liddell 1980). But there is more: if one tells a story in which a whale had initially swallowed a person but has now spit out the person, SVO order is regained—the preferred way to say this is: WHALE SWIM-SPIT-OUT-cl PERSON. This is all the more surprising since the classifier is almost the same as that meaning *swim and swallow*—the manual shape and the movement are the same, but at the end the "mouth" opens instead of closing (Schlenker et al. 2024).

Figure 3. "The whale spit out the person" expressed with the classifier *SWIM-SPIT-OUT-cl*

WHALE



SWIM-SPIT-OUT-cl



PERSON



One proposal to make sense of these facts is that classifiers combine with their arguments in a visual way: when a whale swallows a person, one typically sees the person and the whale before one sees the swallowing, and this is the reason the subject (= WHALE) and object (= PERSON) come before the verb (=SWIM-SWALLOW-cl). By contrast, when a whale thankfully spits out a person it had previously swallowed, one typically sees the whale and the spitting out before one sees the released individual, hence the order SVO (= WHALE SWIM-SPIT-OUT-cl PERSON). On this view, word order is ruled in this case by an iconic syntax that follows that of a very simple comics, not by standard grammar.<sup>5</sup>

In sum, sign language doesn't just have an iconic semantics alongside its standard semantics; it also has an iconic syntax alongside its standard syntax.

#### Lesson 4. Even the purely depictive component of sign has a sophisticated semantic structure.

One might think that the iconic component of sign language classifiers conveys linguistically unstructured pictorial information. But the opposite is true: language has a rich typology of inferential types, and the same typology is found with classifiers (Schlenker et al. 2025).

To illustrate, take the sentence *Sam regrets cheating on the exam*. It's an odd thing to say if it's not taken for granted that Sam cheated, which is the *presupposition* of the sentence. Its main point or *assertion* is that Sam feels bad about cheating. The presupposition and the assertion behave very differently in complex sentences—typically, the assertion is affected by logical words such as *not* and *maybe*, but the presupposition isn't. This effect can be seen in these sentences: *Sam doesn't regret cheating*, *Maybe Sam regrets cheating*, *Does Sam regret cheating?* All take for granted that Sam cheated, but they don't yield an inference that Sam feels bad about it (the negation even denies it). Presuppositions are pervasive in language and are found in multiple further constructions. For instance, *Maybe our plane will take off* presupposes that our plane is currently on the ground, as does the sentence *Our plane won't take off*. This is because *take off* has an "on the ground" presupposition.

Strikingly, classifiers too can generate presuppositions. The classifier for helicopters, which appear in Fig. 4a, can be realized with an upward movement to represent a take-off, as in Fig. 4b (note that this is not a verb meaning "take off"—it is just a classifier going up, and it couldn't represent a plane taking off, for instance). When the same tests as above are performed (with negation, *maybe*, and questions), the conclusion is unmistakable—the rising classifier comes with a presupposition that the helicopter is initially on the ground.

Figure 4.

a. Helicopter classifier    b. Upward movement of a helicopter classifier



<sup>5</sup> There should be interesting connections in the future with linguistically inspired work on comics; see Cohn 2013, Abusch 2020.

If instead of the normal helicopter classifier one uses a representation with two rotors, as in Fig. 5, then one understands that the helicopter has two rotors, and this meaning component too behaves like a presupposition (as various tests have shown).

Figure 5. A classifier representing a two-rotored helicopter



There are many more inferential types in language than just assertions and presuppositions, and to the extent that they have been tested, most appear to exist with classifiers. In sum, the depictive information contributed by classifiers is divided "on the fly" among the slots of the typology of linguistic inferences: it has a sophisticated semantic structure.

### **Lesson 5. Gestures that replace words share non-trivial properties with signs—but unlike signs, they are not words.**

Owing to its rich depictive component, sign may seem to be richer than speech—after all, it has the same type of logical and grammatical operations, but richer means of depiction. In recent years, however, there has been a growing consensus that sign with iconicity should be compared to speech with gestures rather than to speech alone (Goldin-Meadow and Brentari 2017). Strikingly, when gestures are considered, one finds gestural versions of the first four lessons discussed above (Schlenker 2020).

First, some pointing gestures can disambiguate reference in a way that is reminiscent of sign language pointing-based pronouns—a gestural counterpart of Lesson 1. Take the sentence: *Whenever I can hire a mathematician or a sociologist, I pick him*. The reference of *him* is entirely unclear. But with the modification illustrated in Fig. 6, everything becomes clear. *A mathematician* co-occurs with an open right-hand palm, *a sociologist* co-occurs with an open palm left-hand palm, and *him* is fully replaced with a pointing gesture towards the right, hence the meaning *I pick the mathematician*.

Figure 6.



Whenever I can hire    [a mathematician] or    [a sociologist], I pick    ➞-right.

As with the Sarkozy/Obama example discussed above, here positions in gestural space arguably play the same role as logical variables.

Second, gestures clearly provide much iconic information—a gestural counterpart of Lesson 2. For instance, in Fig. 7, a gesture of unscrewing a bulb replaces the verb and provides depictive information about the intended movement, with an extended arm and rotational movement.

Figure 7.



The light bulb in that room, are you going to ?

Third, gestures come with a dedicated syntax, and the details offer a gestural counterpart of Lesson 3. The starting point is a surprising discovery made by Goldin-Meadow et al. 2008: when asked to pantomime scenes in silent gestures, speakers of diverse languages tend to use preverbal objects, even if their native language has postverbal objects, as is the case of English (SVO order). So to say in gestures or pantomimes: "a crocodile swallowed a ball", one could use three gestures in the order CROCODILE BALL EAT. Strikingly, English gestures mirror ASL classifiers in overriding the basic word order of the language, down to the details. If we want to say that "a crocodile spit out a ball", postverbal objects are preferred, hence CROCODILE SPIT-OUT BALL—just as is the case with ASL classifiers. A natural explanation is that like sign language classifiers, spoken language gestures have a special syntax because of their iconic nature—an iconic syntax (Schlenker et al. 2024).

Fourth, the content of gestures is divided "on the fly" among a rich typology of inferential types, just as is the case for classifiers—hence a gestural counterpart of Lesson 4. In the bulb-related example above, the question is whether the addressee will unscrew the bulb. But this takes for granted that the bulb is positioned high: this is a presupposition of the question. This division between assertion and presupposition can be found in many other gestures. For instance, the question in Fig. 8 asks whether at the end of the meeting Ann will remove her glasses, but it presupposes that right before the end of the meeting she will have glasses on. More generally, the informational content of iconic gestures is divided among traditional slots of the inferential typology of language (Schlenker 2019, Tieu et al. 2019).

Figure 8.



At the end of the meeting, will Ann ?

Still, there is an essential difference between gestural contributions in speech and iconic contributions in sign: only the latter can modify the shape of conventional words. Some have a very specific and sometimes abstract meaning, as is the case of *REFLECT*, mentioned above; its meaning would be hard to match in improvised gestures.

## Lesson 6. Gestures that accompany words make backgrounded (non-assertive) contributions.

Most gestures co-occur with spoken words—they are "co-speech gestures", unlike some of the gestures we discussed above, which fully replace some words ("pro-speech gestures"<sup>6</sup>). They introduce interesting complexities on their own, as they are usually backgrounded (Ebert and Ebert 2014)—with important consequences for the comparison between sign with iconicity and speech with gestures.

Initially, it may seem that the sentence with the co-speech gesture in Fig. 9a just means: Asterix will punish his enemy by slapping him.

<sup>6</sup> Intuitively, just like a pronoun "replaces" a noun, a pro-speech gestures replaces a part of speech, hence the terminology. We focused on pro-speech gestures first because, although they are uncommon, they are easier to analyze than co-speech gestures.

Figure 9.



a. Co-speech gesture: His enemy, Asterix will punish .



b. Pro-speech gesture: His enemy, Asterix will .

But in fact the co-speech gesture behaves very differently from the expression "by slapping him" in that its informational contribution is not the main point of the sentence, and for this reason is not targeted by negation, as illustrated in Fig. 10a.

Figure 10.



a. Co-speech gesture: His enemy enemy, Asterix won't punish .

Inference: Asterix won't punish his enemy, but if he did, slapping would be involved.



b. Pro-speech gesture: His enemy, Asterix won't .

The sentence in Fig. 10a means in essence "Asterix won't slap his enemy", but with an assumption that *if he were to punish his enemy, slapping would be involved*. This assumption has been shown to be a presupposition (Schlenker 2018b, Tieu et al. 2018, Esipova 2019), and for this reason the "slapping" component is not targeted by negation. By contrast, *Asterix won't punish his enemy by slapping him* denies the slapping component. The pro-speech gesture in Fig. 9b doesn't involve this complication: it makes an assertive contribution, and is therefore targeted by negation in Fig. 10b.

The upshot is that the most common gesture types in speech, namely co-speech gestures, do not make assertive contributions. This is very different from the modulations of the ASL verb *GROW* (Fig. 1) or of the sign language classifiers discussed above (Fig. 2-5), which typically make an assertive contribution (although they may make other contributions *in addition*, as seen in Fig. 4-5). This suggests that iconic information is often distributed differently in speech and in sign—more often assertive in sign than in speech.<sup>7</sup>

Are there equivalents of co-speech gestures in sign? Signers do sometimes gesture (Emmorey 1999, 2023; see also van Loon et al. 2014). But since the hands are typically occupied by signing, to have a simple counterpart of co-speech gestures, it is expedient to investigate non-grammatical facial expressions. Starting from English, a disgusted co-speech facial expression co-occurring with *go skiing with my parents*, as in Fig. 11, gives rise to a meaning component that is not targeted by negation—just as the co-speech slapping gesture of Fig. 10b. So the meaning obtained is: *I won't go skiing with my parents*, with the presupposition that *if I were to go skiing with them, there would be something unpleasant involved*.

<sup>7</sup> In addition to co-speech and co-sign gestures and facial expressions, recent research has also investigated post-speech and post-sign gestures, which follow the words they modify (rather than co-occurring with them). An example is: My enemy, I will punish — SLAP (where the gesture follows *punish*). Recent research has argued that this construction too involves a non-assertive meaning, but of a different type from what is found with co-speech and co-sign gestures (e.g. Schlenker 2018c).



Figure 11.



I won't go [skiing with my parents].

While sign has several facial expressions that play a key grammatical role (e.g. Pfau and Quer 2010), it also has disgusted facial expressions of this type, and when they co-occur with signs they give rise to the same kind of non-assertive contribution: co-sign facial expressions seem to mirror the behavior of co-speech gestures (Schlenker 2018c).

**Lesson 7. Several of these properties apply beyond signs and gestures, suggesting that they have a deeper cognitive source—not a narrowly linguistic one.**

In several of the cases discussed above, the findings extend to entirely artificial iconic stimuli, suggesting that we are dealing with something broader than language, even construed in multimodal terms. Take the fact that the iconic information contributed by ASL classifiers (Fig. 4-5) and by the "unscrew bulb" gesture (Fig. 7) is divided among an assertion and a presupposition. This finding has been replicated in artificial examples in which a visual animation replaces a verb (for instance to represent an alien changing color from green to blue; Tieu et al. 2019); some examples even involved a musical snippet replacing a verb (for instance an ascending scale to represent a hiker going up a hill; Migotti and Guerrini 2021). There is no way experimental participants could have seen such stimuli in a sentence before (who goes around speaking with visual animations or with tunes inserted in sentences?). So they clearly used a general algorithm to divide informational contents among assertions and presuppositions; and this algorithm is not restricted to linguistic material.

The same point has been made about co-speech gestures, as their behavior can be replicated with co-speech sound effects. For instance, the sound of an explosion co-occurring with *assassinate his target* below gives rise to the presupposition that *if the soldier were to assassinate his target, a bombing would be involved* (Pasternak and Tieu 2022).

Figure 12. A co-speech sound effect.

BOOOOOOOOOOOM

The soldier will not [assassinate his target].

Sound example: <https://youtu.be/W4q5r3hKjvA>

It was even argued that music co-occurring with film or cartoon snippets ("co-film music") displays the same semantic behavior as co-speech gestures, at least when these snippets are inserted in sentences (Schlenker 2022).

The upshot is that several iconic effects are not restricted to linguistic material, and might be explained by more general and potentially non-linguistic algorithms. Finding their cognitive source is one of the most important issues for future research.

## Conclusion

Multimodal studies are significantly reshaping our understanding of meaning. Some findings offer visible confirmation or further applications for abstract mechanisms that had been posited for speech. Others highlight the importance of the gradient, iconic component of language alongside its discrete, logical component. This raises key new questions about their interaction, and suggests an expanded

research program for formal semantics.<sup>8</sup> In several cases, insights from multimodality highlight the existence of powerful algorithms that can apply to new content types, including when these are not linguistic in nature. The investigation of their cognitive source is a key issue for future research, and it should help strengthen the connection between semantics and cognition.

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<sup>8</sup> Despite the wealth of new data and insights offered by multimodal studies, these have been slow to permeate the work of theoretical semanticists. We hope that this will change.

**Figure credits**

Fig. 1, 6, 7, 9, 10, 11: drawings by Marion Bonnet reprinted under Creative Commons CC BY 4.0

Fig. 2 Dr. Bill Vicars, <https://www.lifeprint.com/asl101/pages-signs/classifiers/classifiers-main.htm>

Fig. 3 Schlenker et al. 2024

Fig. 4a Schlenker, Philippe; Lamberton, Jonathan; Lan, Nur; Lamberton, Jason; Geraci, Carlo; Salis, Ambre; Ravaux, Lucie; Grüter, Christoph; Ryder, Robin; Chemla, Emmanuel: to appear, Ancestral Iconicity: The Dance Language of Bees Revisited, *Biological Reviews*.

Fig. 4b, 5., 8 Schlenker, Philippe: 2021, Iconic Presuppositions. *Natural Language & Linguistic Theory* 39:215–289.

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## Further reading

For overviews of sign language semantics, see:

Davidson, K. (in press). *Formal semantics and pragmatics in sign languages*. Cambridge University Press.

Schlenker, P., Lamberton, J., & Kuhn, J. (2024). Sign language semantics. *Stanford Encyclopedia of Philosophy*. <https://plato.stanford.edu/entries/sign-language-semantics/>

For overviews of gesture semantics, see:

Ebert, C. (2024). Semantics of gesture. *Annual Review of Linguistics*, 10(1), 169–189.

Schlenker, P. (in press). The meaning and grammar of pure gestures: Theoretical insights. *Annual Review of Linguistics*.

For a general audience overview of semantics from a multimodal perspective, see:

Schlenker, Philippe: 2022, *What it All Means: Semantics for (Almost) Everything*. MIT Press.